

# Path Metrics Computations & Theories

## 1. Distance Calculation (Spherical & Euclidean Theory)

- 2D Ground Distance (d2d): Uses the Haversine Formula, which is the standard trigonometric theory for calculating the great-circle distance between two points on a sphere (the Earth).

Formula:

$$a = \sin^2(\text{delta\_phi}/2) + \cos(\text{phi1}) * \cos(\text{phi2}) * \sin^2(\text{delta\_lambda}/2)$$

$$c = 2 * \text{atan2}(\text{sqrt}(a), \text{sqrt}(1-a))$$

$$d2d = R * c \text{ (where } R = 6371000 \text{ meters, Earth's radius)}$$

- 3D Travel Distance (d3d): Uses the Pythagorean Theorem to account for elevation changes, ensuring the distance includes the physical hypotenuse of traveling up/down a hill.

$$\text{Formula: } d3d = \text{sqrt}(d2d^2 + dz^2)$$

## 2. Topographical Computations

- Total Ascent/Descent: Loops through every segment between GPX points. If the elevation difference (dz) is positive, it adds it to total\_ascent. If dz is negative, it adds the absolute value to total\_descent.

- Max Slope (max\_slope): Calculated by finding the slope of every individual segment  $((dz / d2d) * 100)$  and tracking the highest absolute value found along the entire route.

- Overall Gradient (gradient): The average slope of the entire path from start to finish.

$$\text{Formula: } (\text{elev\_diff} / \text{total\_dist\_2d}) * 100$$

## 3. Kinematic Penalty Theory (Speed & Travel Time)

Instead of a complex non-linear curve (like Tobler's function), this script uses a Linear Slope Penalty Theory:

- Base Speed: 8.33 m/s (roughly 30 km/h, typically implying vehicle travel).

- Slope Penalty: Linear rule where base speed reduces by 2% for every 1% of absolute overall gradient. Capped at a maximum penalty of 80% (0.8) to ensure theoretical movement on steep hills.

$$\text{Formula: } \text{slope\_penalty} = \min(\text{abs}(\text{gradient}) * 0.02, 0.8)$$

- Adjusted Speed (adj\_speed\_mps):

$$\text{Formula: } \text{base\_speed\_mps} * (1.0 - \text{slope\_penalty})$$

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- Travel Time (tt\_slope\_s):

Formula:  $\text{Time} = \text{Distance (3D)} / \text{Adjusted Speed}$ .

It is then divided by 60 to compute the 'Travel\_Time\_with\_Slope\_min' used in path calculations.